Food Management Systems

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Food Management Systems (A36342)

Short Title: Food Management Systems
Faculty Science and Computing
Department Science
Cluster Food Science
Author JFRISBY
Credits 5 **Level:** Intermediate

Description of Module / Aims

The module is designed to give students knowledge and understanding of the various quality systems that are used in the food industry, including aspects of teamwork, benchmarking and problem-solving techniques. The module also prepares the students to have the skills to implement a HACCP plan in the food industry.

Programmes

AGRI-0094	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	3 / 5 / M

Indicative Content

- Total Quality Management: quality improvement programmes, customer satisfaction, benchmarking
- Quality auditing: types of audits, how to conduct a compliance audit
- Problem-solving techniques and their uses: pareto analysis, cause and effect analysis, brainstorming
- Food hazards: microbiological, chemical, physical
- Application of GHP and GMP as a pre-requisite to HACCP
- Examination of the key components of risk analysis, namely risk assessment, risk management and risk communication
- HACCP system and its implementation
- ISO22000, BRC etc.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Examine the various Food Safety Management Systems such as HACCP, BRC and ISO22000.
2. Classify the various types of quality audits.
3. Appraise the importance of GHP and GMP as a pre-requisite to HACCP.
4. Develop a HACCP plan for the food industry.
5. Investigate the various hazards and risks in the food chain.
6. Use techniques for critical thinking and decision making.

Learning and Teaching Methods

- Lectures.
- Case studies.
- Site visit.
- Self-study of assigned reading.
- Guest speaker.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	50%	3, 4, 5, 6
Assignment	10%	5
In-Class Assessment	40%	1, 2, 3, 5, 6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Essential Material

- "Module, Food Management Systems." www.moodle.wit.ie
- Motarjemi, Y. and H. Lelieveld. _Food Safety Management: a practical guide for the food industry._. United States of America: Amsterdam: Academic Press, 2014.

Supplementary Material

- "Food Safety Authority of Ireland." www.fsai.ie
- Goetsch, D.L. and S. Davis. _Quality Management: introduction to quality management for production, processing and services_. United States: Prentice Hall, 2003.

- Kolarik, W.J. _Creating Quality-Concepts, Systems, Strategies and Tools_. London: McGraw-Hill, 1995.
- Lal, H. _Organisational Excellence through Total Quality Management: a practical approach_. New Dehli: New Age International Ltd, 2008.
- Motarjemi, Y. _Hygiene in Food Processing_. U.K.: Academic Press, 2014.